

Contribution:

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SimpleLoader Implementation:

We pass the ELF file into launch, which then calls the `load_and_run_elf` function. This function first opens the file in read-only mode. Next we allocate memory for an `Elf32_Ehdr` struct and read the ELF Header from the file into it. And we get the values of some attributes to allocate memory for the program headers. And we seek the `e_phoff` in the file (phdr table offset). Next we read all the program headers into memory. And we iterate through the table to check if `p_type` is "PT_LOAD" and if it is we check whether the entrypoint is inside the segment and if it is we allocate memory for that segment using `mmap` and read the content of the segment into memory. Then we calculate the Entrypoint and jump into the code image.

We used Makefiles to compile in each directory. And added one top makefile which calls all the make commands, and also moves files into the bin directory as given in the assignment.

The user simply has to go to main directory and call `make`, after which he/she should go to bin directory, and type `./launch ../test/fib` to get the output.

Github link: <https://github.com/jazpeak/Custom-ELF-Loader>